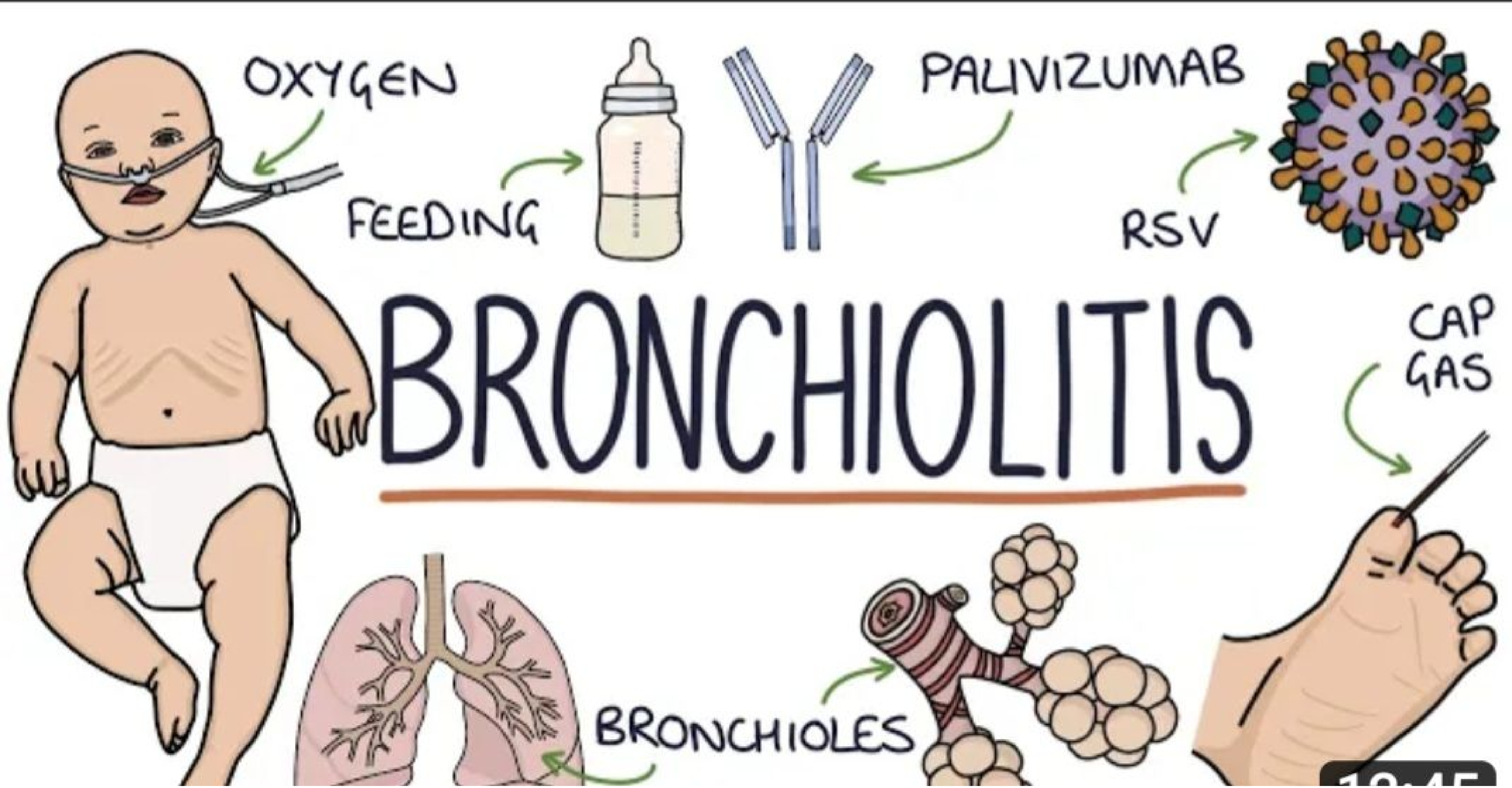
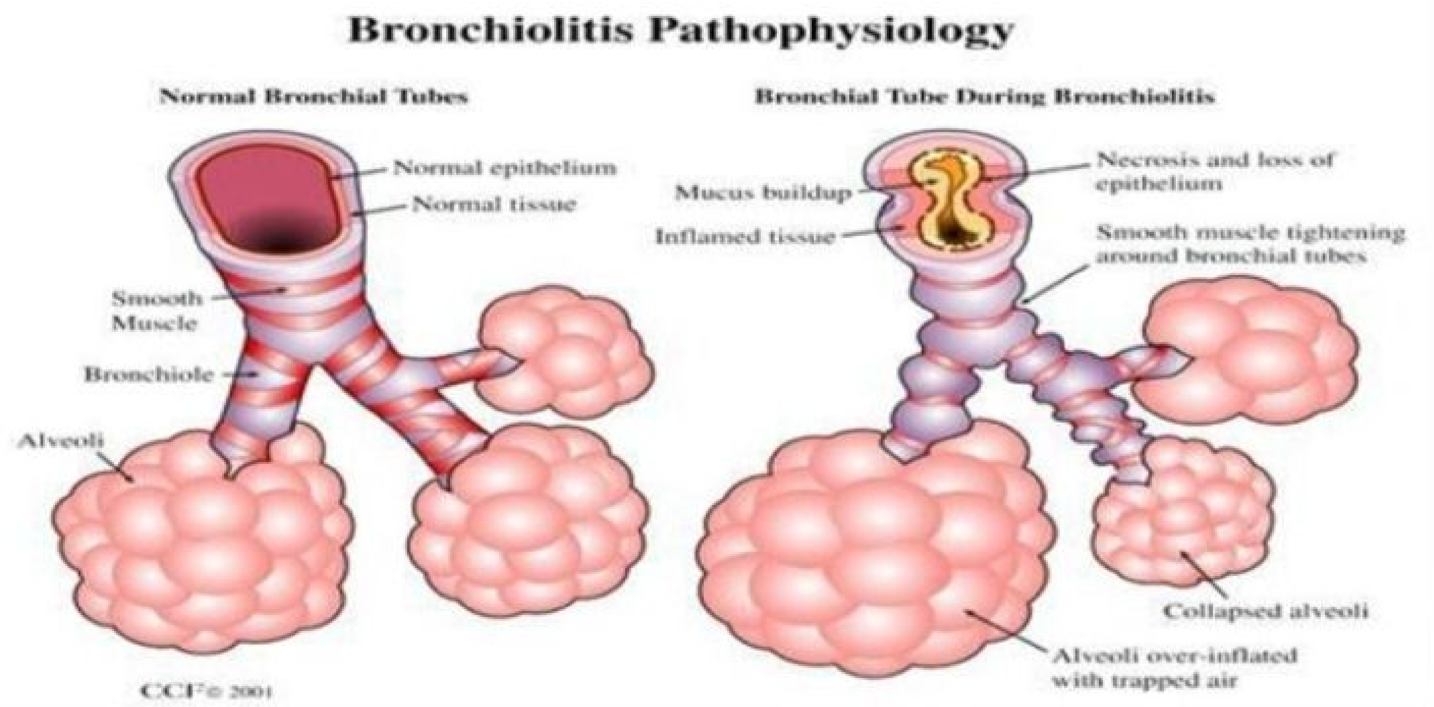

OXYGEN
FEEDING
PALIVIZUMAB
RSV
BRONCHIOLITIS
CAP
GAS
BRONCHIOLES

DEFINITION

- **Bronchiolitis**
  - A viral lower respiratory tract infection characterized by obstruction of small airways caused by acute inflammation, edema and necrosis of the epithelial cells lining the small airways as well as increased mucus production

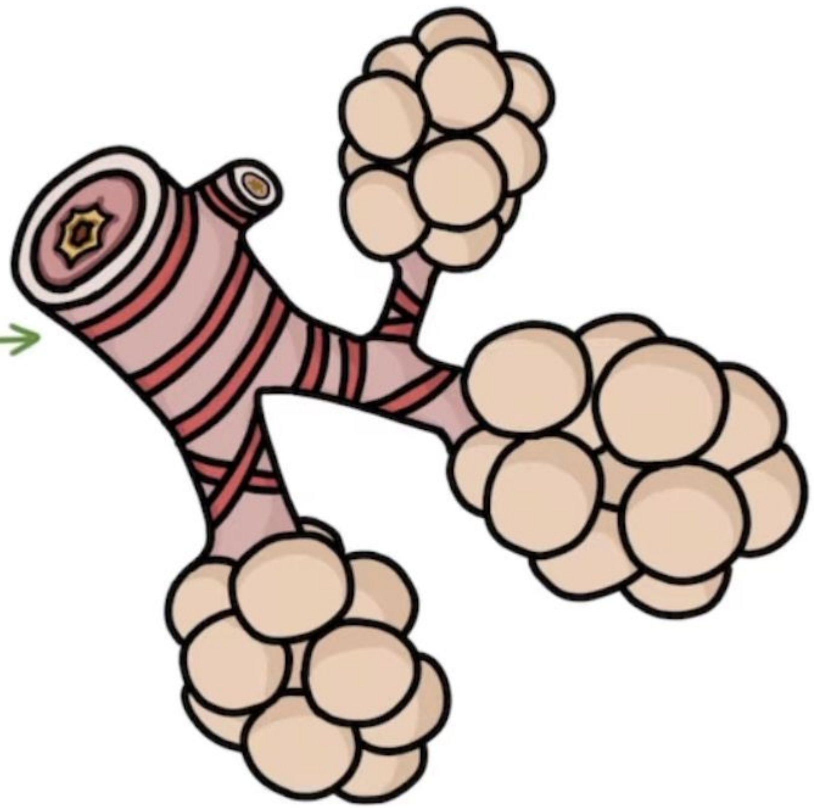


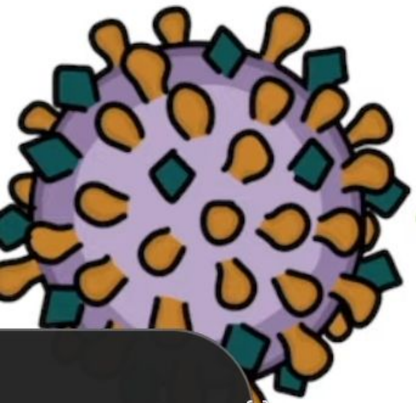
BRONCHIOLITIS → COMMON IN WINTER!

→ INFLAMMATION

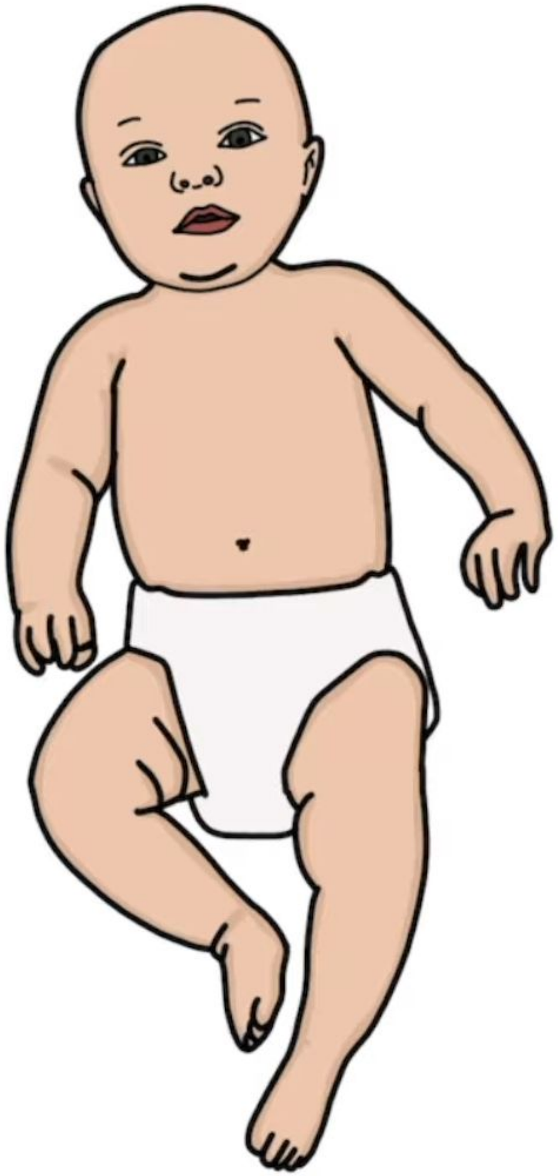
→ INFECTION

BRONCHIOLES





RESPIRATORY SYNCYTIAL VIRUS (RSV)

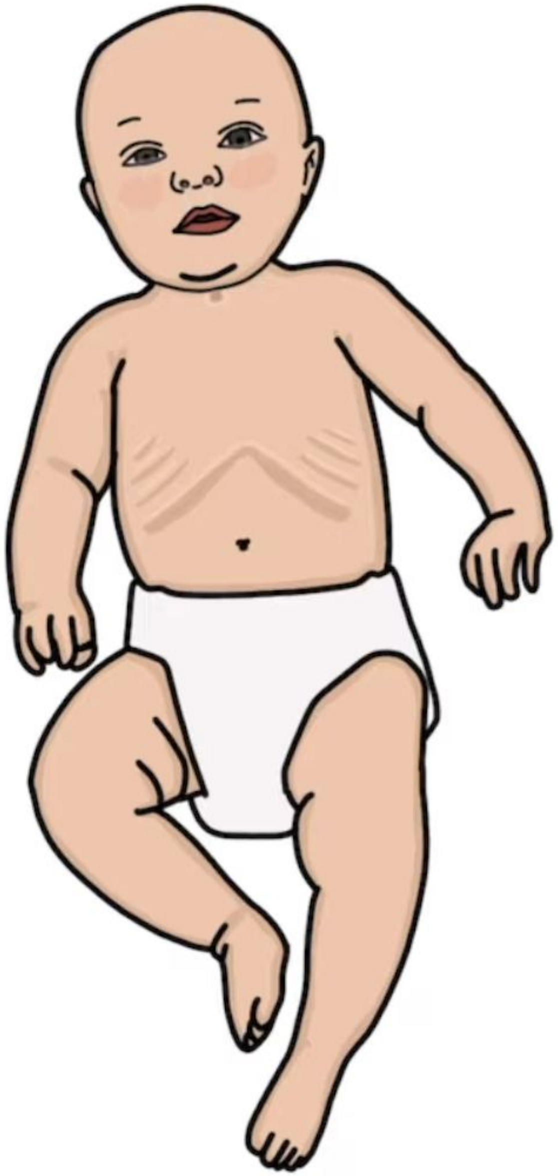


BRONCHIOLITIS

OCCURS IN:

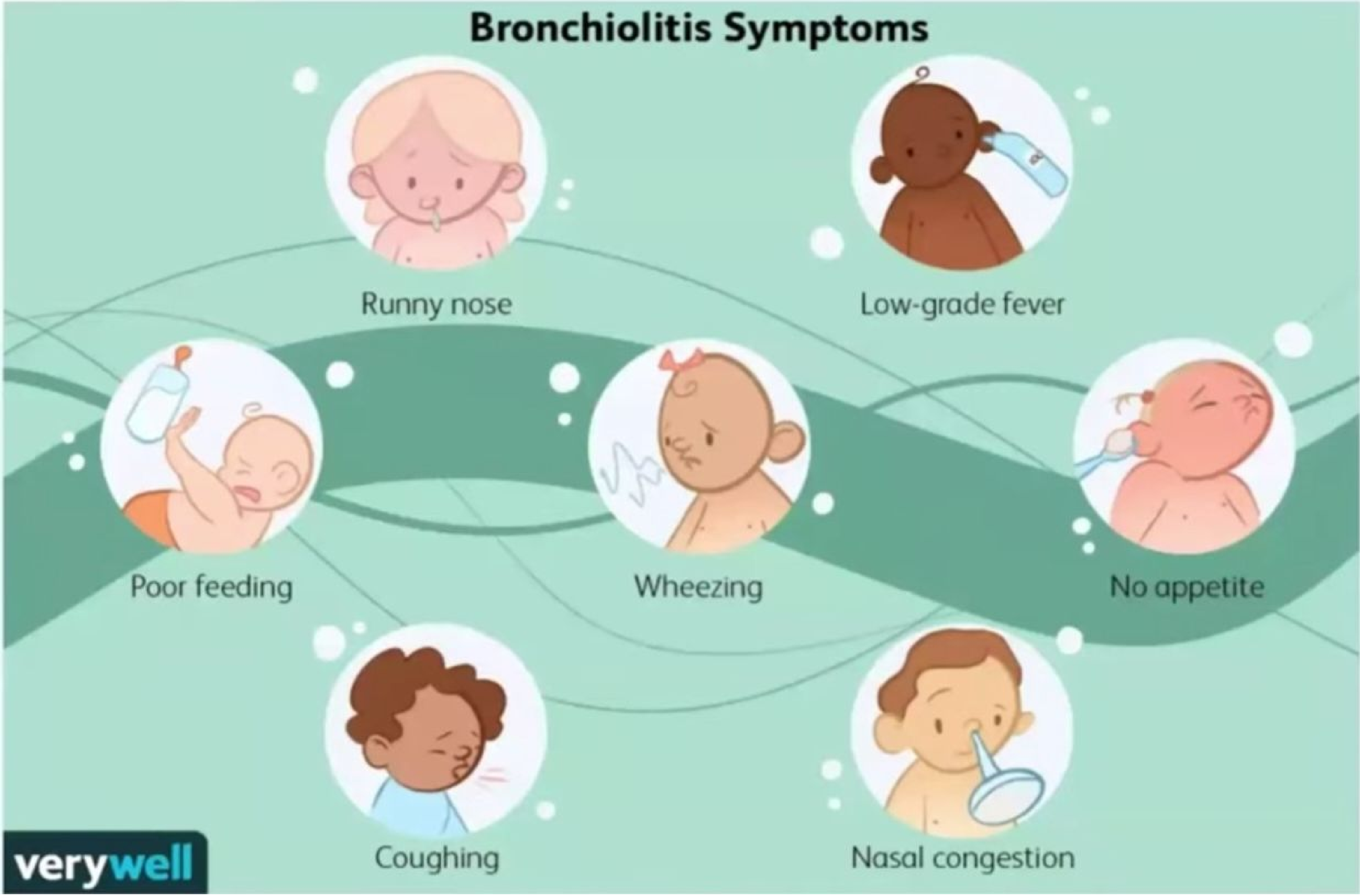
- UNDER 1 YEAR
- MOST COMMON UNDER 6 MONTHS
- RARELY UP TO 2 YEARS
  - EX-PREMATURE
  - CHRONIC LUNG DISEASE

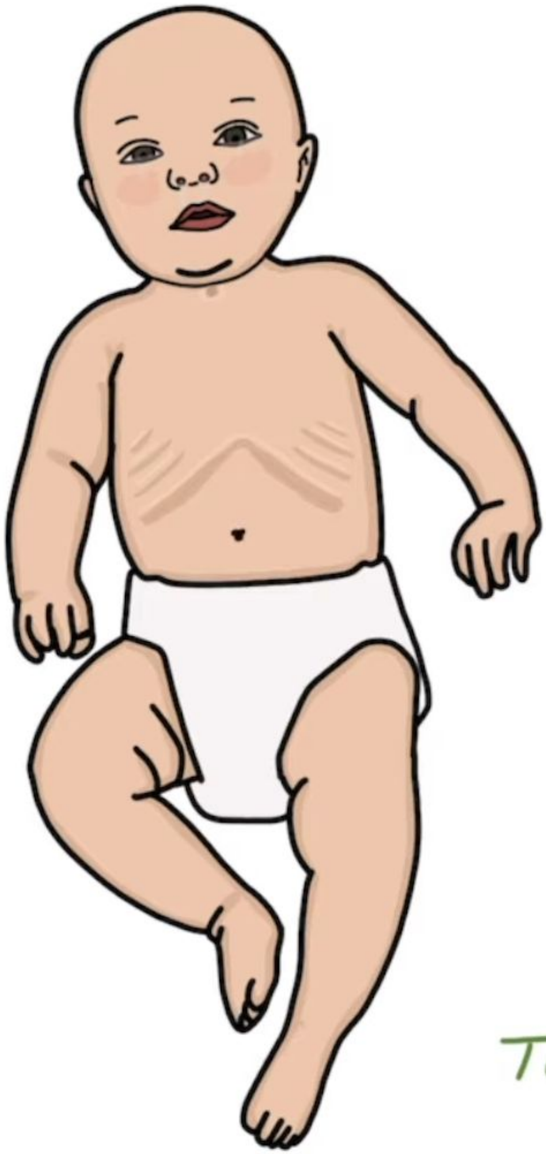
- CORYZAL SYMPTOMS
  - RUNNY / SNOTTY NOSE
  - SNEEZING
  - MUCUS IN THROAT
  - WATERY EYES



PRESENTATION

- CORYZAL SYMPTOMS
- SIGNS OF RESPIRATORY DISTRESS
- DYSPNOEA
- TACHYPNOEA
- POOR FEEDING
- MILD FEVER → UNDER 39°C
- APNOEAS
- WHEEZES
- CRACKLES


Bronchiolitis Symptoms
Runny nose
Low-grade fever
Poor feeding
Wheezing
No appetite
Coughing
Nasal congestion
verywell



SIGNS OF RESPIRATORY DISTRESS

- → ↑ RESPIRATORY RATE
- → USE OF ACCESSORY MUSCLES
- → INTERCOSTAL RECESSIONS
- → SUBCOSTAL RECESSIONS
- → NASAL FLARING
- → HEAD BOBBING
- → TRACHEAL TUGGING
- → CYANOSIS
- → ABNORMAL AIRWAY NOISES

TOM TIP: BECOME VERY CONFINENT IN LISTING AND SPOTTING THE SIGNS

ABNORMAL AIRWAY NOISES

WHEEZING — WHISTLING SOUND — NARROW AIRWAYS — EXPIRATION

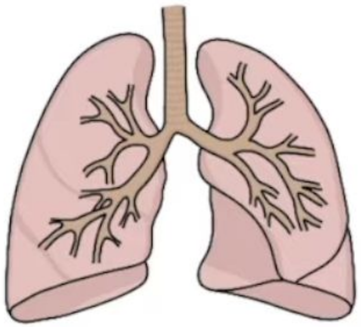
GRUNTING — EXHALING WITH GLOTTIS PARTIALLY CLOSED — INCREASE POSITIVE END-EXPIRATORY PRESSURE

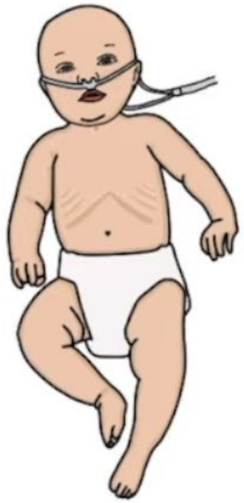
STRIDOR — HIGH PITCHED INSPIRATORY — OBSTRUCTION UPPER AIRWAY — E.G. CROUP

TYPICAL COURSE

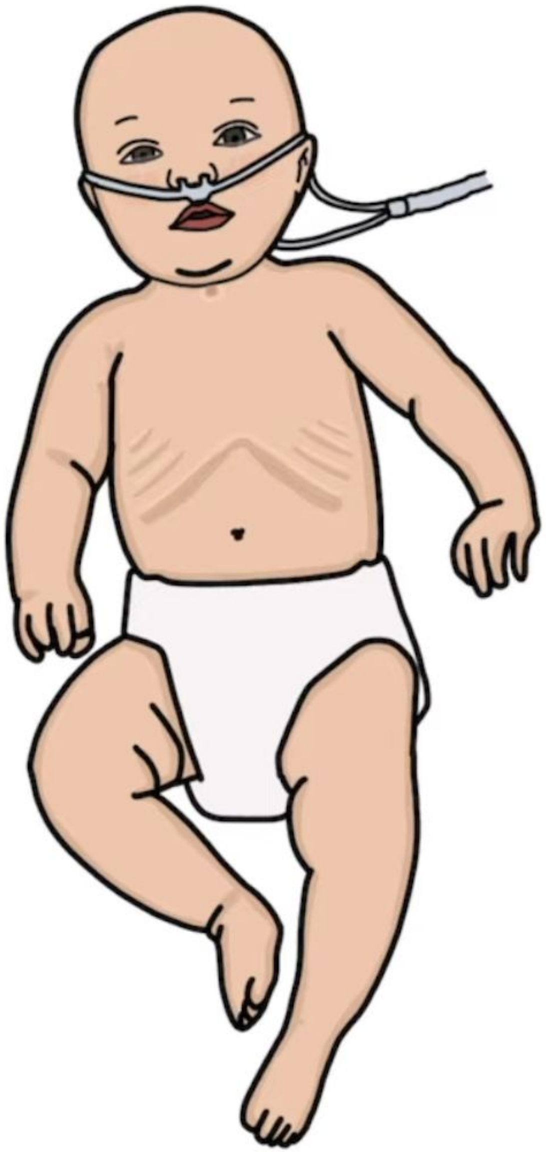
BRONCHIOLITIS AS INFANT

↳ VIRAL-INDUCED WHEEZE





START	1-2 DAYS	3-4 DAYS	7-10 DAYS	2-3 WEEKS
CORYZAL SYMPTOMS	CHEST SYMPTOMS	SYMPTOMS WORST	SYMPTOMS LAST	FULLY RECOVER

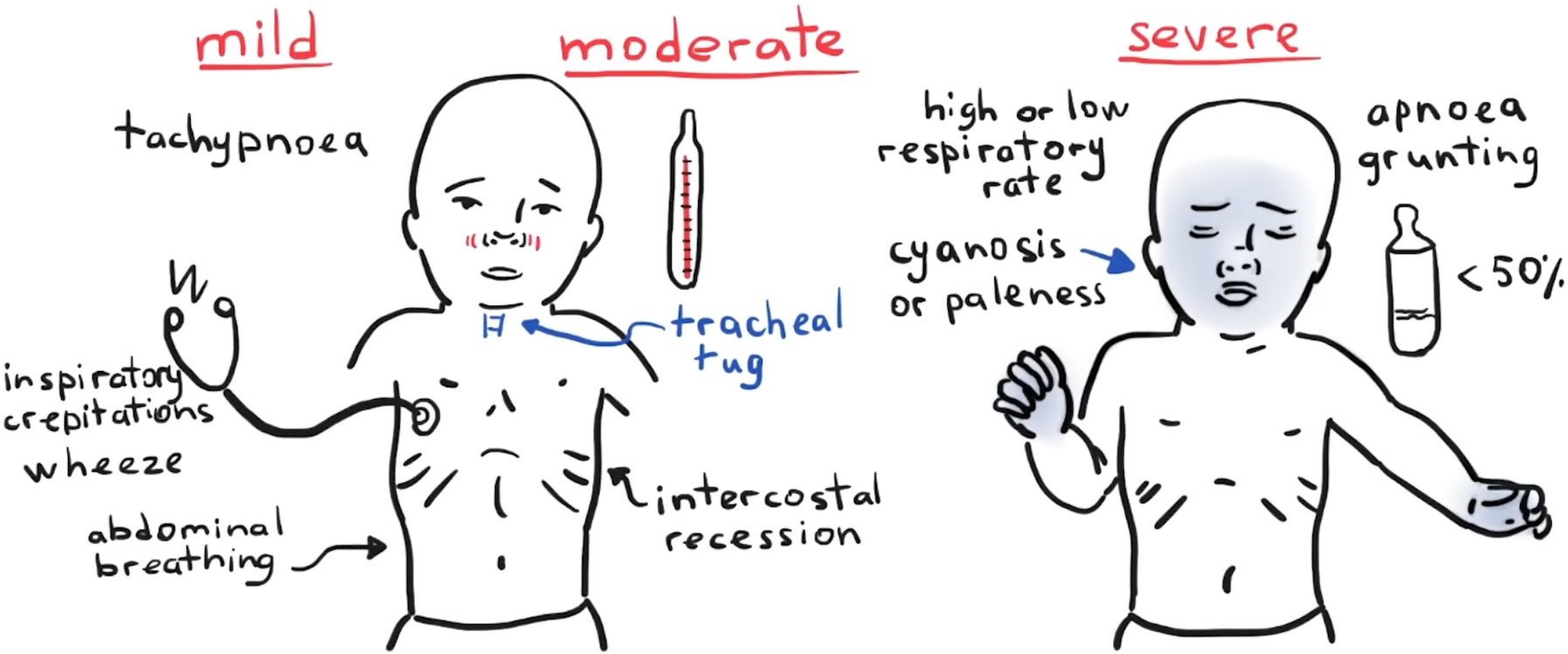


CRITERIA FOR ADMISSION

- AGED UNDER 3 MONTHS
- PRE-EXISTING CONDITION
- < 50-75% OF NORMAL INTAKE

30%

- CLINICALLY DEHYDRATED
- RESPIRATORY RATE >70
- OXYGEN SATURATIONS <92%
- MODERATE-SEVERE RESPIRATORY DISTRESS
- APNOEAS
- PARENTS CONCERNED

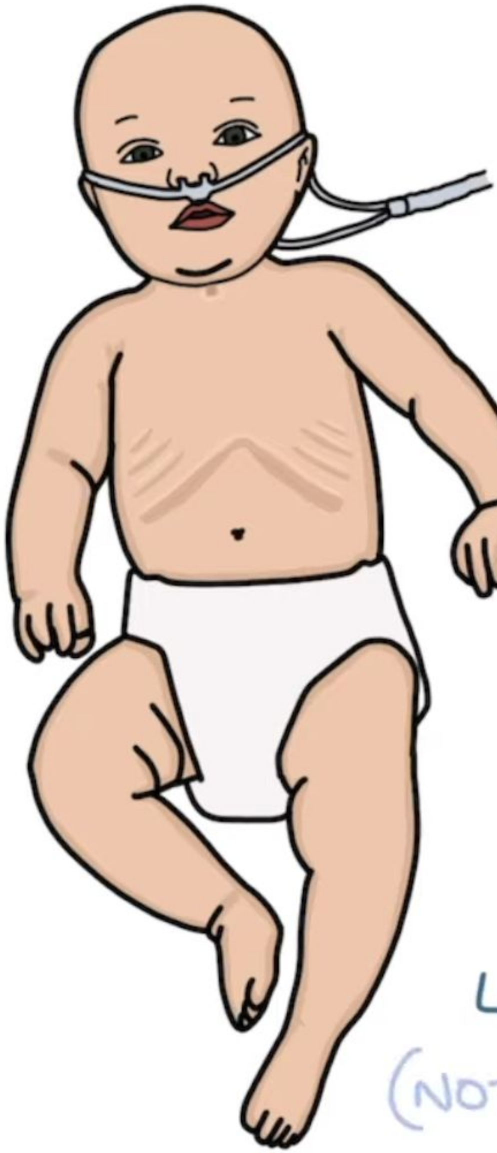


Differential diagnosis

Acute asthma Viral induced wheeze Pneumonia
Congestive heart failure Pertussis

Risk factors

<6 weeks
premature or lower weight for gestation
immunodeficiency
congenital heart disease
neurological conditions
chronic respiratory illness



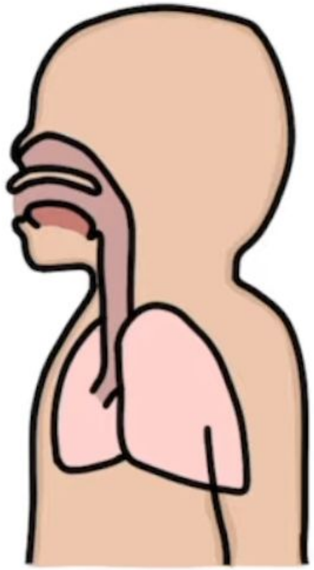
MANAGEMENT → SUPPORTIVE

- ENSURE ADEQUATE INTAKE
- SALINE NASAL DROPS
- NASAL SUCTION
- SUPPLEMENTARY OXYGEN → SATS >92%
- VENTILATORY SUPPORT

LITTLE EVIDENCE FOR
(NOT ROUTINELY USED)

- NEBULISED SALINE
- BRONCHODILATORS
- STEROIDS
- ANTIBIOTICS

VENTILATORY SUPPORT → STEPPED UP



1. HIGH FLOW HUMIDIFIED OXYGEN
2. CONTINUOUS POSITIVE AIRWAY PRESSURE
3. INTUBATION AND VENTILATION

ASSESSING VENTILATION

CAPILLARY BLOOD GAS

- SEVERE RESPIRATORY DISTRESS
- MONITORING VENTILATORY SUPPORT

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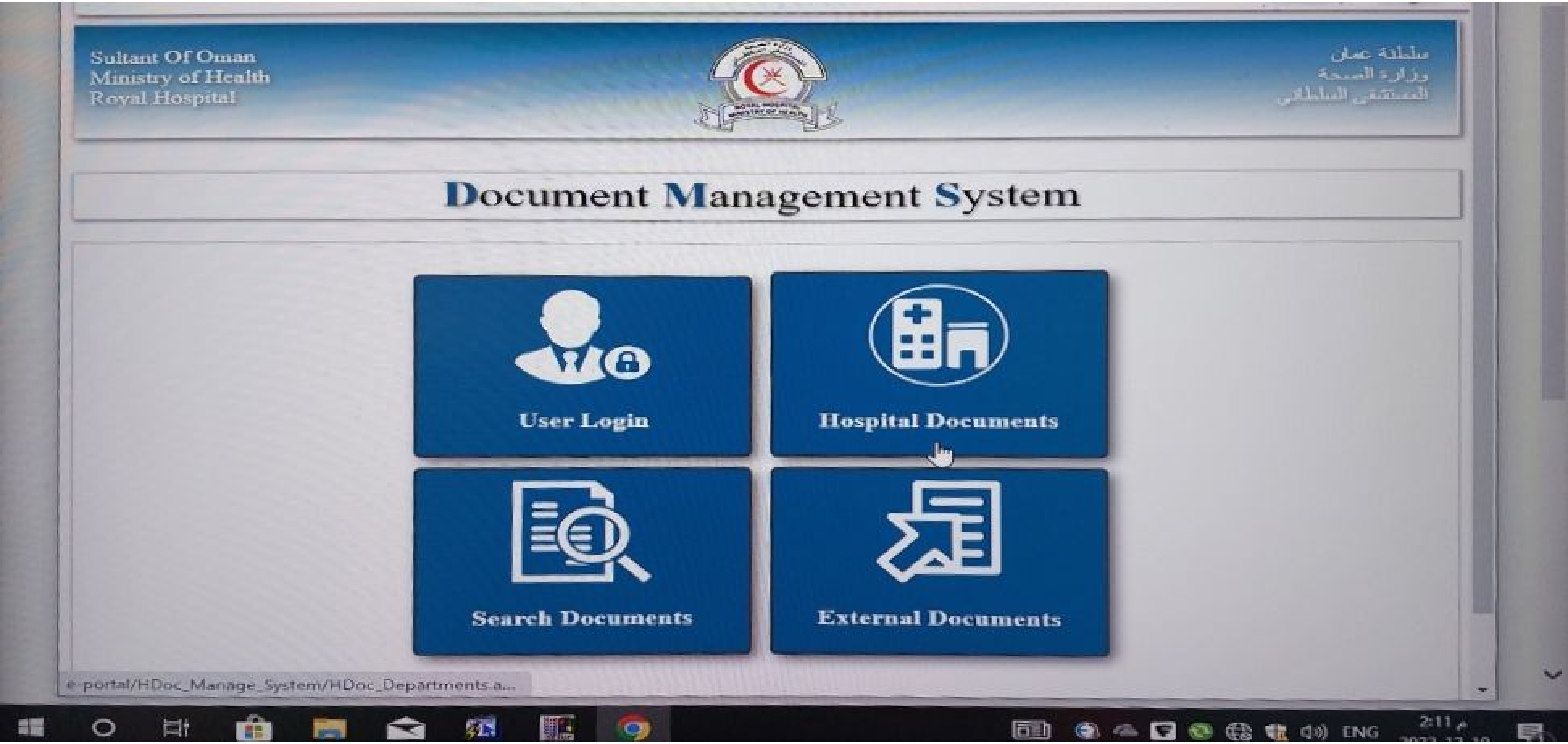
Clinical Job Description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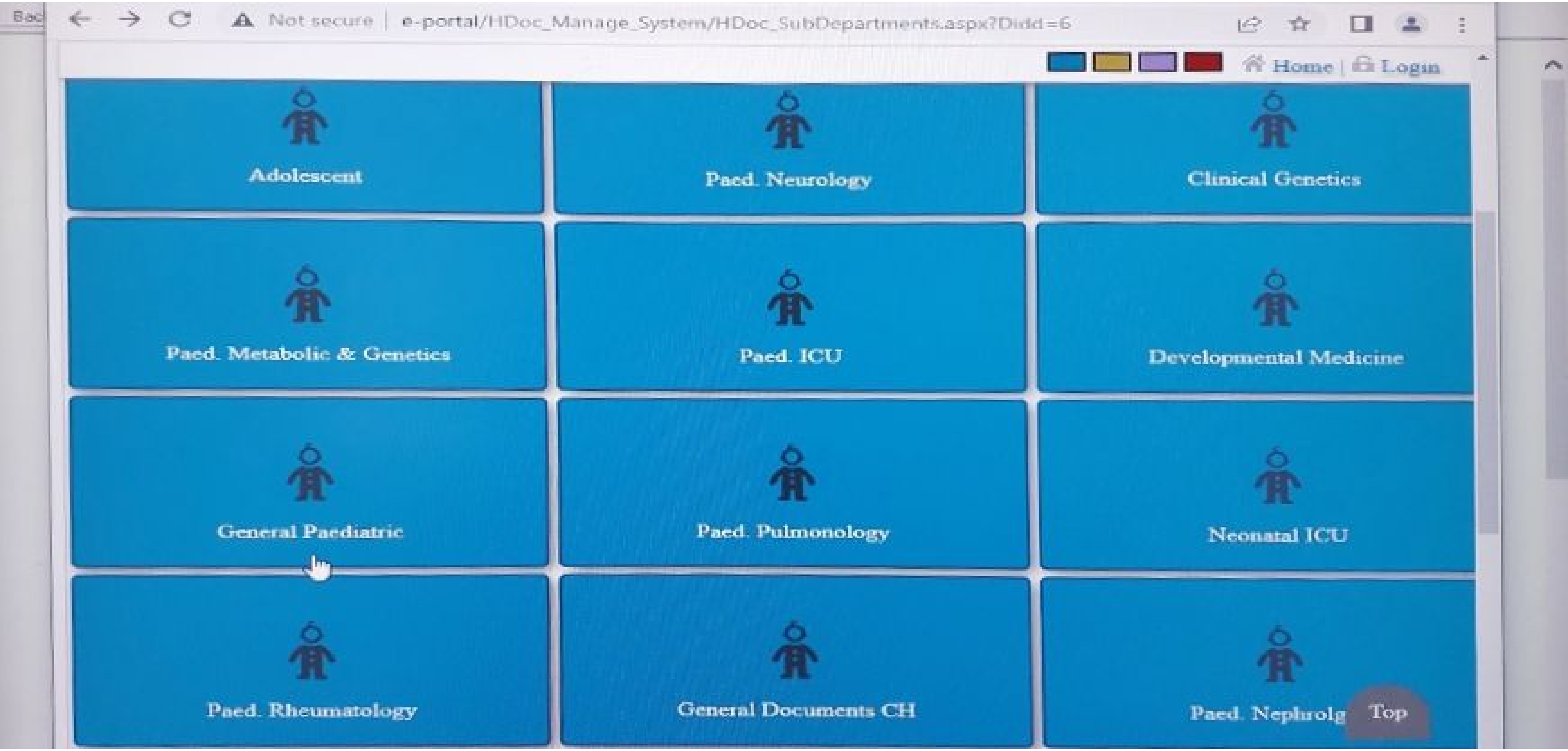
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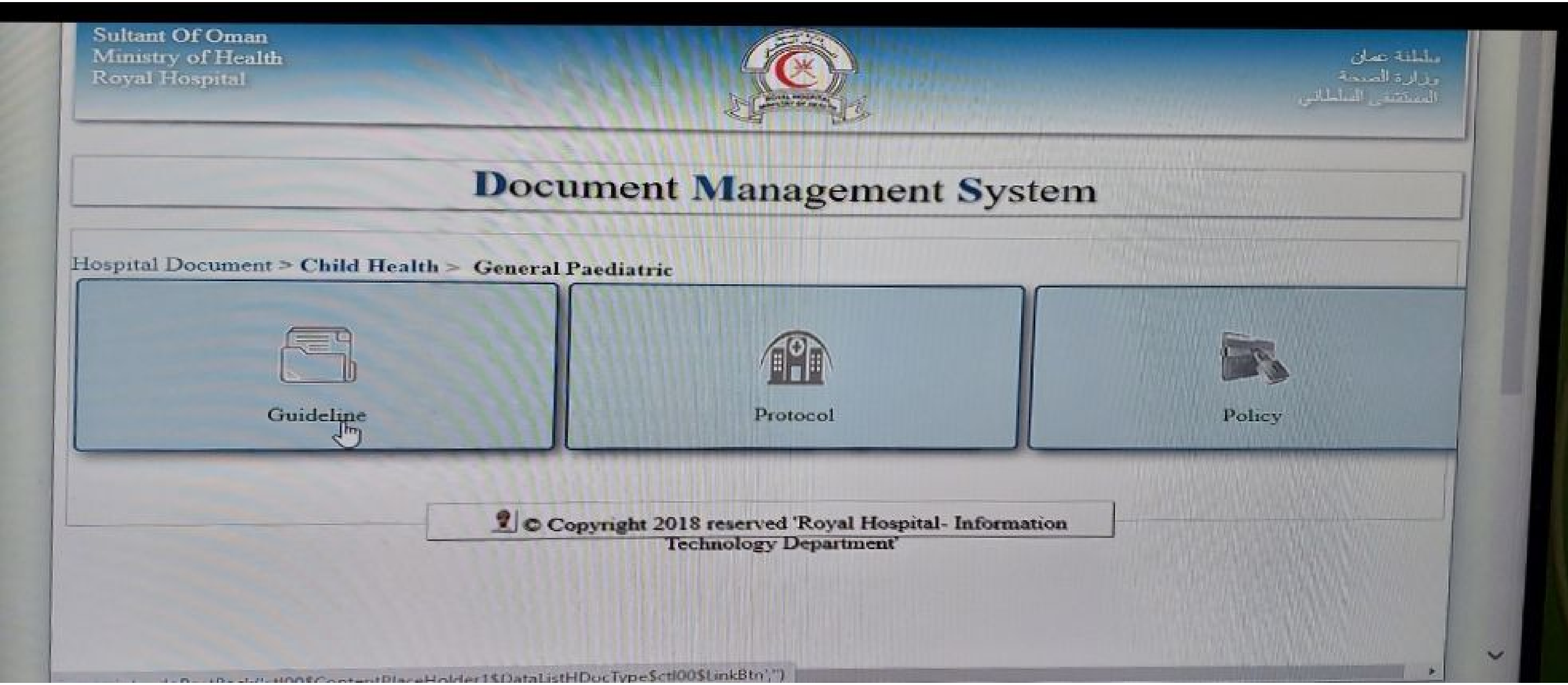
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	Title
1	Approach to Osteopetrosis in Paediatric Age Group
2	Acute Bronchiolitis
3	Pathway For Difficult Pediatric IV Access During Working Hours
4	Pathway for Difficult Pediatric IV access during on call time
5	Accidental Tracheostomy Dislodgement Algorithm
6	Evaluation and Management of Well-Appearing Febrile Infants 0 to 60 Days Old
7	Intravenous Fluid Therapy
8	Suspected child maltreatment or violation of a childs right (Expired)
9	Guideline of the management of central venous catheters related blockage for children (Expired)
10	Sepsis & Septic Shock Management (Expired)
11	Bed Utilization in the Paediatric Wards (Expired)

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Title: Acute Bronchiolitis

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Acute Bronchiolitis
Recommendations for Diagnosis, Monitoring and Management
For otherwise well children between one-24 months of age

Acute bronchiolitis is the most common reason for admissions to the hospital in the first year of life. There is tremendous variation in the clinical management of this condition around the world, including significant use of unnecessary tests and ineffective therapies.

Purpose and Aim:

This guideline has been developed to provide an evidence-based clinical framework for the management of infants (1-24) months with acute bronchiolitis treated in Pediatric Emergency room or General Pediatric wards.

- These recommendations are intended to decrease the use of unnecessary diagnostic studies and ineffective medications and interventions.
- These recommendations Do not apply to children admitted to the intensive care unit (PICU) or those with chronic lung disease, immunodeficiency and other serious underlying chronic conditions.

Diagnosis:

Viral Bronchiolitis is a clinical diagnosis, based on typical history and examination. Peak severity is usually at around day two to three of the illness with resolution over 7-10 days. The cough may persist for weeks. Bronchiolitis most commonly occurs in the winter months, but can be seen all year around.

Features:

Bronchiolitis typically begins with an acute upper respiratory tract infection followed by onset of respiratory distress and fever and one or more of the following:

- Cough
- Tachypnea
- Retractions, nasal flaring
- Widespread crackles or wheeze
- Grunting, color change or Apnea
- Lower O2 saturations

Evidence-based reviews have NOT supported the use of diagnostic testing in typical cases of

Top

with chronic lung disease, immunodeficiency and other serious underlying chronic conditions.

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Viral Bronchiolitis is a clinical diagnosis, based on typical history and examination. Peak severity is usually at around day two to three of the illness with resolution over 7-10 days. The cough may persist for weeks. Bronchiolitis most commonly occurs in the winter months, but can be seen all year around.

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- Widespread crackles or wheeze
- Grunting, color change or Apnea
- Lower O2 saturations

u Evidence-based reviews have NOT supported the use of diagnostic testing in typical cases of bronchiolitis. (refer to table: 1)

u Tests are often unhelpful and can lead to unnecessary admissions, further testing and ineffective therapies.

Admission indications may include:

- Signs of severe respiratory distress (e.g: indrawing, grunting, tachypnea)
- Supplemental O2 required to keep saturations >90%
- Dehydration or history of poor fluid intake.
- Cyanosis or history of apnea.
- Infant at high risk for severe disease. (refer to table: 2)
- Family unable to cope.

Top

- Infant at high risk for severe disease. (refer to table: 2)
- Family unable to cope.

Table: 2- Risk factors for more serious illness
○ Gestational age less than 37 weeks.
○ Chronological age at presentation less than 10 weeks.
○ Post-natal exposure to cigarette smoke.
○ Breast fed for less than two months.
○ Failure t thrive
○ Chronic lung disease.
○ Congenital heart disease.
○ Chronic neurological conditions.
Infants with any of these risk factors are more likely to deteriorate rapidly and require escalation of care. Consider hospital admission even if presenting early in illness with mild symptoms.

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- Supplemental O2 required to keep saturations >90%
- Dehydration or history of poor fluid intake.
- Cyanosis or history of apnea.
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- Family unable to cope.

Table: 3 - Therapies for Treatment of acute bronchiolitis

- Family unable to cope.

Table: 3 - Therapies for Treatment of acute bronchiolitis (For more detailed explanations, please refer to appendex2)		
Recommended by evidence	**Evidence Equivocal**	**Not recommended**
Oxygen	Epinephrine nebulization	Salbutamol
Hydration	Nasal suctioning	Corticosteroids Antibiotics Antivirals 3% hypertonic saline nebulization Chest physiotherapy Cool mist therapies or therapy with saline aerosol

Monitoring:

- Thoughtful use of oxygen saturation monitoring in hospitalized patients is recommended.
- Continuous saturation monitoring may be indicated for high-risk children in the acute phase of illness, and intermittent monitoring or spot checks are appropriate for lower-risk children and patients who are improving clinically.

References:

1- Canadian Pediatric Society, Acute bronchiolitis position statement, updated Nov 2021.
2- Australian Bronchiolitis guideline.
3- America Academy of pediatrics, Acute bronchiolitis, updated 2014.

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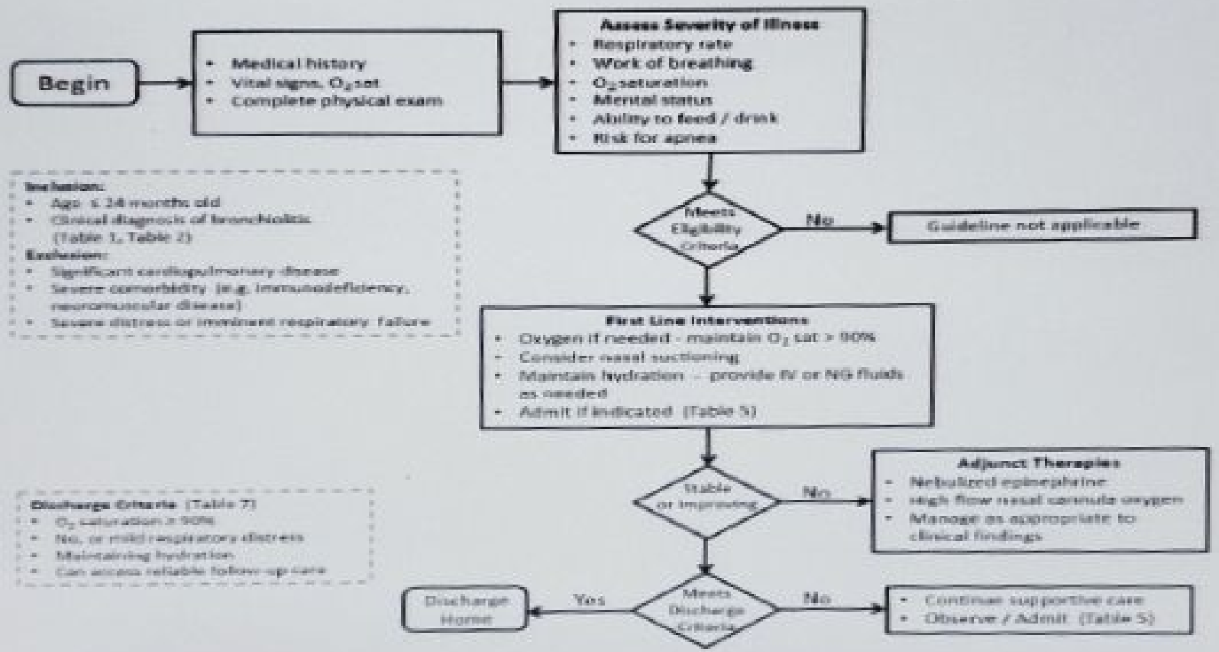

algorithm

Appendix – 2

v Therapies recommended based on evidence:

1-Oxygen:

algorithm
Appendix – 2
v Therapies recommended based on evidence:
1-Oxygen:
o Supplemental oxygen therapy is a mainstay of treatment in hospital. Oxygen should be administered if saturations fall below 90% and used to maintain saturations at ≥90%
o To minimize handling, oxygen is usually administered via nasal cannula, face mask or a head box.
o A recent alternative is heated, humidified, high-flow nasal cannula (HHHFNC) oxygen therapy. There is increasing evidence this mode of
o non-invasive respiratory support is safe, and may reduce the need for CPAP and mechanical ventilation in moderate and severe bronchiolitis.
o Currently, there is insufficient evidence to support its routine use in bronchiolitis.
2- Hydration:
o Some degree of fluid supplementation is required in 30% of hospitalized patients with bronchiolitis.
o Frequent feeds should be encouraged and breastfeeding supported.
o Infants with a respiratory rate >60 breaths/min, particularly those with nasal congestion, may have an increased risk of aspiration and may not be safe to feed orally.
o When supplemental fluids are required, a recent randomized trial found nasogastric (NG) and intravenous (IV) routes to be equally effective, with no difference in length of hospital stay.
o If NG bolus feeds are not tolerated, slow continuous feeds are an option.
o If the IV route is used, isotonic fluids (0.9% NaCl/5% dextrose) are preferred for maintenance, with regular monitoring of serum Na because of the risk of hyponatremia.
v Therapies for which the evidence is equivocal:
1- Nebulized epinephrine:
o Some studies have shown that epinephrine nebulization may be effective for reducing hospital admissions. However, the evidence remains insufficient to support routine use of epinephrine in the emergency department.
o It may be reasonable to administer a dose of epinephrine and carefully monitor clinical response; however, unless there is clear evidence of improvement, continued use is not appropriate.

1- Nebulized epinephrine:

o Some studies have shown that epinephrine nebulization may be effective for reducing hospital admissions. However, the evidence remains insufficient to support routine use of epinephrine in the emergency department.

o It may be reasonable to administer a dose of epinephrine and carefully monitor clinical response; however, unless there is clear evidence of improvement, continued use is not appropriate.

o A systematic review of 19 studies evaluating the use of epinephrine in bronchiolitis shows no effect on length of hospital stay and there is insufficient evidence to support its routine use in admitted patients.

2- Nasal suctioning:

o As for many long-standing and commonly used therapies for children, there is scant evidence supporting the use of nasal suctioning in the management of bronchiolitis. While it appears that suctioning mucus out of blocked nares would be a harmless procedure, one recent study has suggested that deep suctioning and long intervals between suctioning are associated with increased length of stay.

o This suggests that if suctioning is performed, it should be done superficially and reasonably frequently.

v Therapies not recommended based on evidence:

1- Salbutamol:

o Children with bronchiolitis present with a wheeze that is clinically similar to that observed with asthma. However, the pathophysiology of bronchiolitis is such that the airways are obstructed rather than constricted.

o Furthermore, infants appear to have inadequate β-agonist lung receptor sites and immature bronchiolar smooth muscles.

o While studies have shown small improvements in clinical scores, bronchodilators have not been shown to improve O2 saturation, do not reduce admission rates and do not shorten the duration of stay in hospital.

o When the diagnosis of bronchiolitis is clear, a trial of salbutamol is NOT recommended.

2- 3% Hypertonic saline nebulization:

o It has been hypothesized that hypertonic saline increases mucociliary clearance and rehydrates airway surface liquid, with questions as to impact on inpatient length of stay.

hypertonic saline increases mucociliary clearance and rehydrates airway surface liquid, with questions as to impact on inpatient length of stay.

o Recent evidence from meta-analysis, which accounted for important variation among included studies, does not support the use of hypertonic saline.

o In a typical North American population, 3% hypertonic saline has no impact on length of stay and should not be used routinely.

3- Corticosteroids:

o Corticosteroids, such as dexamethasone, prednisone or inhaled glucocorticoids, are not associated with a clinically significant improvement in disease, as measured by reduction in clinical scores, rates of hospitalization and length of hospital stay.

o Furthermore, any small benefit that corticosteroids may offer must be weighed against the risks of steroid treatment.

o Corticosteroids are not recommended for routine use in bronchiolitis.

4- Antibiotics:

o Many children with acute bronchiolitis are prescribed an antibiotic. However, bacterial infection in otherwise healthy children with bronchiolitis is exceedingly rare.

o Research on the role of antibiotics in bronchiolitis is limited and has, to date, failed to identify any benefit.

o Further research is needed to develop criteria for identifying the minority of patients at high risk for secondary bacterial infection.

o Currently, antibiotics should not be used except in cases in which there is clear evidence or strong suspicion of a secondary bacterial infection.

5- Antiviral:

o Antiviral therapies, such as ribavirin, are expensive, cumbersome to administer, provide limited benefit and are potentially toxic to care providers and, thus, are not recommended for the routine treatment of bronchiolitis in otherwise healthy children.

o In patients with or at risk for particularly severe disease, antivirals could be considered, but this decision should be made on an individual basis in consultation with appropriate subspecialists.

6- Chest physiotherapy:

o Nine clinical trials comparing physiotherapy with no treatment were reviewed.

suspicion of a secondary bacterial infection.

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o Antiviral therapies, such as ribavirin, are expensive, cumbersome to administer, provide limited benefit and are potentially toxic to care providers and, thus, are not recommended for the routine treatment of bronchiolitis in otherwise healthy children.

o In patients with or at risk for particularly severe disease, antivirals could be considered, but this decision should be made on an individual basis in consultation with appropriate subspecialists.

6- Chest physiotherapy:

o Nine clinical trials comparing physiotherapy with no treatment were reviewed.

o Neither vibration and percussion nor passive expiratory techniques were shown to improve clinical scores or to reduce hospital stay or duration of symptoms.

o Chest physiotherapy is not recommended for the treatment of

7- Cool mist therapies or aerosol therapy with isotonic saline:

o Cool mist and other aerosol therapies have been used for some time to manage bronchiolitis, with scant evidence supporting their efficacy.

o A recent Cochrane review concluded that there is no evidence supporting or refuting the use of cool mist and other aerosols for managing bronchiolitis.

Written by: Dr.Israa Sadeq & Dr.Aza AlSawafi
Checked by: Dr.Aza AlSawafi
Authorized by: SAMI SULAIMAN SAIF AL FARSI

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Thank
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